Sampling Strategies

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What is Sampling? Why is it Important?

- **Sampling** is the process of selecting a **subset** of a population to collect data from, instead of studying the entire population.

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Why Do We Use Sampling?

- Saves time and resources .
- Makes research feasible.
- Ensures accuracy.

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Two Main Types of Sampling Methods

- Probability Sampling
- Non-Probability Sampling

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Probability Sampling:

- Every individual in the population has an **equal** chance of being selected.
- **Minimizes bias** and provides **accurate** results.

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Simple Random Sampling (SRS)

- Every member of the population is assigned a number.
- A random number generator selects individuals.

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Stratified Sampling

- The population is divided into **subgroups** based on characteristics .
- A **random sample** is taken from each subgroup.

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Cluster Sampling

- The population is divided into **clusters**.
- A few **entire clusters** are **randomly chosen**, and all members of those clusters are surveyed.

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Non-Probability Sampling

- **Selection is not random** some members **have a higher chance** of being selected than others.
- Used when **quick results** are needed, but **less accurate** than probability sampling.

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Convenience Sampling

- The researcher selects participants who are **easiest to reach**.

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Purposive Sampling

- The researcher selects participants **based on specific criteria**.
- **Example:**
  A **health researcher** wants opinions from **only diabetic patients**—so they select only those with diabetes.
- **Advantages:**
  - Good for **specialized research**
  - Ensures **only relevant people** are included
- **Disadvantages:**
  - **Not generalizable** to a large population

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Snowball Sampling

- Researchers start with a **small group** of participants, who then recruit more participants.
- **Example:**
  A study on **drug addiction** starts with **one person** who introduces the researcher to others with similar experiences.
- **Advantages:**
  - Useful for **hard-to-reach populations**
  - Can **expand the sample quickly**
- **Disadvantages:**
  - **High bias risk**
  - Difficult to **verify sample diversity**

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When to Use Each?

- Use **probability sampling** when **accuracy matters**.
- Use **non-probability sampling** when **time & cost are limited**.

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Why is probability sampling more accurate than non-probability sampling?

- Random Selection Reduces Bias

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Why is probability sampling more accurate than non-probability sampling?

- Random Selection Reduces Bias
- Higher Representativeness of the Population
- Allows for Statistical Inference & Generalization

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Determining Sample Size

- The sample size determines how **accurate and reliable** the survey results will be.
  - If the sample is **too small**, the results may not be **representative** of the population.
  - If the sample is **too large**, it **wastes resources** without improving accuracy significantly.

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Why do we calculate sample size?

- Ensures results are **statistically valid**
- Reduces **sampling error**
- Increases **confidence** in conclusions
- Helps balance **cost vs. accuracy**

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Key Factors Affecting Sample Size

- **Population Size (N)**
  - The **total number** of individuals in the group we are studying.
  - If the population is **small**, we may need a **larger proportion** of it in the sample.
  - If the population is **very large**, the required sample size does **not increase proportionally**.

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Key Factors Affecting Sample Size

- **Margin of Error (E):**
  - Also called **sampling error**, it measures how much the sample result **may differ** from the actual population value.
  - A **smaller margin of error** means **more accurate results** but requires a **larger sample**.
  - Common margins of error: **1%, 3%, 5%**

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Key Factors Affecting Sample Size

- **Confidence Level (Z-Score)**
  - Measures how **certain** we are that the sample reflects the population.
  - A higher confidence level means a **larger sample is needed**.
  - Common confidence levels and their **Z-scores**:
    - **90% Confidence Level → 1.645**
    - **95% Confidence Level → 1.96** (Most common)
    - **99% Confidence Level → 2.576**

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Key Factors Affecting Sample Size

- **Standard Deviation (σ)**
  - Measures **how much variation** exists in the population.
  - If responses vary widely, a **larger sample** is needed.
  - If responses are similar, a **smaller sample** is enough.
  - If we **don't know** the population standard deviation, we **assume σ = 0.5** .

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Sample Size Formula

- **$N = z^2 * p * (1-p)/E^2$**

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Sample Size Formula

- **Example:**

A researcher wants to survey **university students** about online learning satisfaction.

- Confidence Level = **95%** → Z = **1.96**
- Estimated Proportion (p) = **0.5**
- Margin of Error (E) = **5%**

- **Solution :**
  - **$N=(1.96)^2*0.5*(1-0.5)/(0.05)^2 =$**

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Adjusting Sample Size for Small Populations

- $n_{adjusted}$ = n/1+n-1/N

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Adjusting Sample Size for Small Populations

- **Example:**
  - If the university **only has 2,000 students**
- **Solution:**
  - $N_{adjusted}$ = 384/1+384-1/2000 = **322**

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Why Does a Larger Sample Size Increase Accuracy?

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- Reduces Sampling Error

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- Reduces Sampling Error
- Decreases Margin of Error

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Why Does a Larger Sample Size Increase Accuracy?

- Reduces Sampling Error
- Decreases Margin of Error
- Improves Confidence Level

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Why Does a Larger Sample Size Increase Accuracy?

- Reduces Sampling Error
- Decreases Margin of Error
- Improves Confidence Level
- Reduces the Effect of Outliers

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